

h	$::=$ emph	M	
t	$::=$ skip A $t; t'$ $t t'$ $atomic(B)\{t\}[P]$ $while(B)\{t\}$ (t)	M	commands skip Abstract Command sequencing parallel composition atomic operation while loop
wf_t	$::=$ skip A $wf_t; wf_t'$ $wf_t wf_t'$ $atomic(B)\{no_atomic_t\}[P]$ $while(B)\{wf_t\}$ (wf_t)	M	commands skip Abstract Command sequencing parallel composition atomic operation while loop
no_atomic_t	$::=$ skip A $no_atomic_t; no_atomic_t'$ $no_atomic_t no_atomic_t'$ $while(B)\{no_atomic_t\}$ (no_atomic_t)	M	commands skip Abstract Command sequencing parallel composition while loop
l	$::=$ R p		
n	$::=$ $n + 1$ 0		
s	$::=$ (h_1, h_2, h_3)		
A	$::=$ s	M	abstract command
B	$::=$		abstract boolean
P	$::=$		abstract boolean
R, G	$::=$ Emp	M	interference

<i>Config</i>	::=	configuration
		(t, s)
		fault
<i>formula</i>	::=	
		$A(h_1, h_2)$
		$R(h_1, h_2)$
		$h_1 * h_2 = h_3$
		h_1 notin A
		disjoint (h_1, h_2, h_3)
		$B(h)$
		$P(h)$
		$B(h)$
		B fault on h
		$h / = P_1 * \mathbf{true}$
		<i>judgement</i>
<i>Jop</i>	::=	
		$Config_1 \xrightarrow{l} Config_2$ $Config_1$ reduces to $Config_2$
		$Config_1 \xrightarrow{1}^* Config_2$
<i>judgement</i>	::=	
		<i>Jop</i>
<i>user_syntax</i>	::=	
		h
		t
		wf_t
		no_atomic_t
		l
		n
		s
		A
		B
		P
		R
		<i>Config</i>
		<i>formula</i>

1 Look, see

Here's the original rendering:

$$\frac{\displaystyle \frac{Config_1 \xrightarrow{R} Config_2}{Config_2 \xrightarrow{1}^* Config_3}}{Config_1 \xrightarrow{1}^* Config_3} \text{ ENV}$$

And here's Matt's LaTeX hackery in use (potentially):

$$\begin{array}{c}
\frac{\frac{Config_1 \xrightarrow{R} Config_2}{Config_2 \xrightarrow{1}^* Config_3}}{Config_1 \xrightarrow{1}^* Config_3} \quad \text{ENV} \\
\\
\frac{\frac{B(h) \quad h' \mid / = P_1 * \mathbf{true} \quad h_1 * h_2 = h}{(t, (h, \mathbf{emph}, h_3)) \xrightarrow{1}^* (\mathbf{skip}, (h', \mathbf{emph}, h'_3))}}{(atomic(B)\{t\}[P_1], (h_1, h_2, h_3)) \xrightarrow{P} \mathbf{fault}} \quad \text{FAULT_ATOMIC_EXIT} \\
\\
\boxed{Config_1 \xrightarrow{l} Config_2} \quad Config_1 \text{ reduces to } Config_2 \\
\\
\frac{}{(\mathbf{skip}; t, s) \xrightarrow{P} (t, s)} \quad \text{SKIP_SEQ} \\
\\
\frac{}{(t, s) \xrightarrow{P} (t_1, s')} \\
\\
\frac{}{(t; t', s) \xrightarrow{P} (t_1; t', s')} \quad \text{SEQ} \\
\\
\frac{}{(\mathbf{skip} \parallel t, s) \xrightarrow{P} (t, s)} \quad \text{SKIP_PAR_L} \\
\\
\frac{}{(t \parallel \mathbf{skip}, s) \xrightarrow{P} (t, s)} \quad \text{SKIP_PAR_R} \\
\\
\frac{}{(t_1, s) \xrightarrow{P} (t'_1, s')} \\
\\
\frac{}{(t_1 \parallel t_2, s) \xrightarrow{P} (t'_1 \parallel t_2, s')} \quad \text{PAR_L} \\
\\
\frac{}{(t_2, s) \xrightarrow{P} (t'_2, s')} \\
\\
\frac{}{(t_1 \parallel t_2, s) \xrightarrow{P} (t_1 \parallel t'_2, s')} \quad \text{PAR_R} \\
\\
A(h_1, h'_1) \\
\mathbf{disjoint}(h'_1, h_2, h_3) \\
\\
\frac{}{(A, (h_1, h_2, h_3)) \xrightarrow{P} (\mathbf{skip}, (h'_1, h_2, h_3))} \quad \text{PROGRAM_ACTION} \\
\\
\frac{\begin{array}{l} P_1(h'_2) \\ B(h) \\ h'_1 * h'_2 = h' \\ h_1 * h_2 = h \end{array}}{(t, (h, \mathbf{emph}, h_3)) \xrightarrow{1}^* (\mathbf{skip}, (h', \mathbf{emph}, h'_3))} \\
\\
\frac{}{(atomic(B)\{t\}[P_1], (h_1, h_2, h_3)) \xrightarrow{P} (\mathbf{skip}, (h'_1, h'_2, h'_3))} \quad \text{ATOMIC} \\
\\
\frac{\begin{array}{l} B(h) \\ h' \mid / = P_1 * \mathbf{true} \\ h_1 * h_2 = h \end{array}}{(t, (h, \mathbf{emph}, h_3)) \xrightarrow{1}^* (\mathbf{skip}, (h', \mathbf{emph}, h'_3))} \\
\\
\frac{}{(atomic(B)\{t\}[P_1], (h_1, h_2, h_3)) \xrightarrow{P} \mathbf{fault}} \quad \text{FAULT_ATOMIC_EXIT}
\end{array}$$

$$\begin{array}{c}
B(h) \\
h_1 * h_2 = h \\
(t, (h, \mathbf{emph}, h_3)) \xrightarrow{1}^* \mathbf{fault}
\end{array}$$

$$\frac{}{(atomic(B)\{t\}[P_1], (h_1, h_2, h_3)) \xrightarrow{P} \mathbf{fault}} \quad \text{FAULT_ATOMIC_BODY}$$

$$\begin{array}{c}
B \mathbf{fault\ on\ } h \\
h_1 * h_2 = h
\end{array}$$

$$\frac{}{(atomic(B)\{t\}[P_1], (h_1, h_2, h_3)) \xrightarrow{P} \mathbf{fault}} \quad \text{FAULT_ATOMIC_GUARD}$$

$$\begin{array}{c}
B(h_1)
\end{array}$$

$$\frac{}{(while(B)\{t\}, (h_1, h_2, h_3)) \xrightarrow{P} (t; while(B)\{t\}, (h_1, h_2, h_3))} \quad \text{WHILE_TRUE}$$

$$\begin{array}{c}
B(h_1)
\end{array}$$

$$\frac{}{(while(B)\{t\}, (h_1, h_2, h_3)) \xrightarrow{P} (\mathbf{skip}, (h_1, h_2, h_3))} \quad \text{WHILE_FALSE}$$

$$\begin{array}{c}
B \mathbf{fault\ on\ } h_1
\end{array}$$

$$\frac{}{(while(B)\{t\}, (h_1, h_2, h_3)) \xrightarrow{P} \mathbf{fault}} \quad \text{WHILE_FAULT}$$

$$\begin{array}{c}
h_1 \mathbf{notin\ } A
\end{array}$$

$$\frac{}{(A, (h_1, h_2, h_3)) \xrightarrow{P} \mathbf{fault}} \quad \text{FAULT_PRIM}$$

$$\begin{array}{c}
(t, s) \xrightarrow{P} \mathbf{fault}
\end{array}$$

$$\frac{}{(t; t_2, s) \xrightarrow{P} \mathbf{fault}} \quad \text{FAULT_SEQ}$$

$$\begin{array}{c}
(t, s) \xrightarrow{P} \mathbf{fault}
\end{array}$$

$$\frac{}{(t || t_2, s) \xrightarrow{P} \mathbf{fault}} \quad \text{FAULT_PAR_L}$$

$$\begin{array}{c}
(t, s) \xrightarrow{P} \mathbf{fault}
\end{array}$$

$$\frac{}{(t_2 || t, s) \xrightarrow{P} \mathbf{fault}} \quad \text{FAULT_PAR_R}$$

$$\begin{array}{c}
R(h_2, h'_2) \\
\mathbf{disjoint\ } (h_1, h'_2, h'_3)
\end{array}$$

$$\frac{}{(t, (h_1, h_2, h_3)) \xrightarrow{R} (t, (h_1, h'_2, h'_3))} \quad \text{ENVIRONMENT_ACTION}$$

$$\boxed{Config_1 \xrightarrow{1}^* Config_2}$$

$$\frac{}{Config_1 \xrightarrow{1}^* Config_1} \quad \text{REFL}$$

$$\begin{array}{c}
Config_1 \xrightarrow{R} Config_2 \\
Config_2 \xrightarrow{1}^* Config_3
\end{array}$$

$$\frac{}{Config_1 \xrightarrow{1}^* Config_3} \quad \text{ENV}$$

$$\begin{array}{c}
Config_1 \xrightarrow{\mathbf{p}} Config_2 \\
Config_2 \xrightarrow{\mathbf{l}}^* Config_3 \\
\hline
Config_1 \xrightarrow{\mathbf{l}}^* Config_3 \quad \text{PROG}
\end{array}$$

Definition rules: 22 good 0 bad
Definition rule clauses: 54 good 0 bad